

SPB Leg Building Block, Rev A: Fabrication Board

Gate driver, isolated supplies, DC link, parasitics, connectors and fabrication data

Enes Ayaz, KTH — design note for the $2 \times \text{EPC2361}$ per switch building block

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Summary.

- One half-bridge leg (two mirrored commutation cells, $2 \times \text{EPC2361}$ per switch) on a 4-layer, 2 oz board of 34×85.4 mm with DC terminals at one end, the AC terminals directly below the switch nodes, the gate supplies and the lab header on the tab. The layout is generated by `scripts/make_kicad_bb_fab.py`; DRC: 0 violations, 0 unconnected pads. A KiCad schematic generated from the same netlist (`spb-bb-fab.kicad.sch`) matches the board pin by pin.
- **Gate drive:** one 2EDF7275K for both switches, two isolated supplies (one per channel), centre star per switch with Kelvin returns. Separate turn-on and turn-off paths per channel: R_{ON} in parallel with a Schottky diode in series with R_{OFF} , then $R_{G,i}$ per transistor. Default $R_{\text{ON}} = 0.75 \Omega$, $R_{\text{OFF}} = 0 \Omega$, $R_{G,i} = 0.5 \Omega$: 2.0Ω (on) and 0.5Ω plus diode (off) external per transistor. Split rail $+5.05/-1.24$ V from MGN1S1208MC $\rightarrow$ LP2951 (6.29 V) $\rightarrow$ TLV431. Peak gate current 3.1 A (on) / 6.5 A (off) per channel, below the 4/8 A driver rating.
- **Gate loops in the corridor:** the high-side trunk runs on L3 with its Kelvin return stacked directly below on L4; the low-side star is on L1 with the Kelvin collector on L2. Both Kelvin collectors join on the symmetry line, so the two transistors of a switch see identical gate loops.
- **DC link:** $2 \times 6 \times 1 \mu\text{F}$ 0805 at the drains plus $24 \times 10 \mu\text{F}$ 1210 bank, unchanged from the paper design (Table 4).
- **Before ordering:** Q3D re-extraction of this routed copper (26 of 168 power vias were removed for clearance and the 0805 pitch went from 1.4 to 1.6 mm), check of the LP2951 adjustable-mode pins and the TLV431 pinout, and an isolation review for the 1200 V stack (the 2EDF7275K is a functional-isolation part).

1 Board overview and stack-up

Fig. 1 shows the four copper layers. From top to bottom: the DC terminals J1/J2 (DC+, DC–), the three MLCC bank rows, the local 0805 capacitors, the two mirrored cells with the high-side (row at $y = 4.9$ mm) and low-side (row at $y = 9.2$ mm) transistors, the driver in the 9 mm corridor between the cells, the two AC terminals J3/J4 directly below the switch nodes, the two gate-supply chains, the two isolated DC/DC modules and, at the bottom, the primary side with the lab header J5.

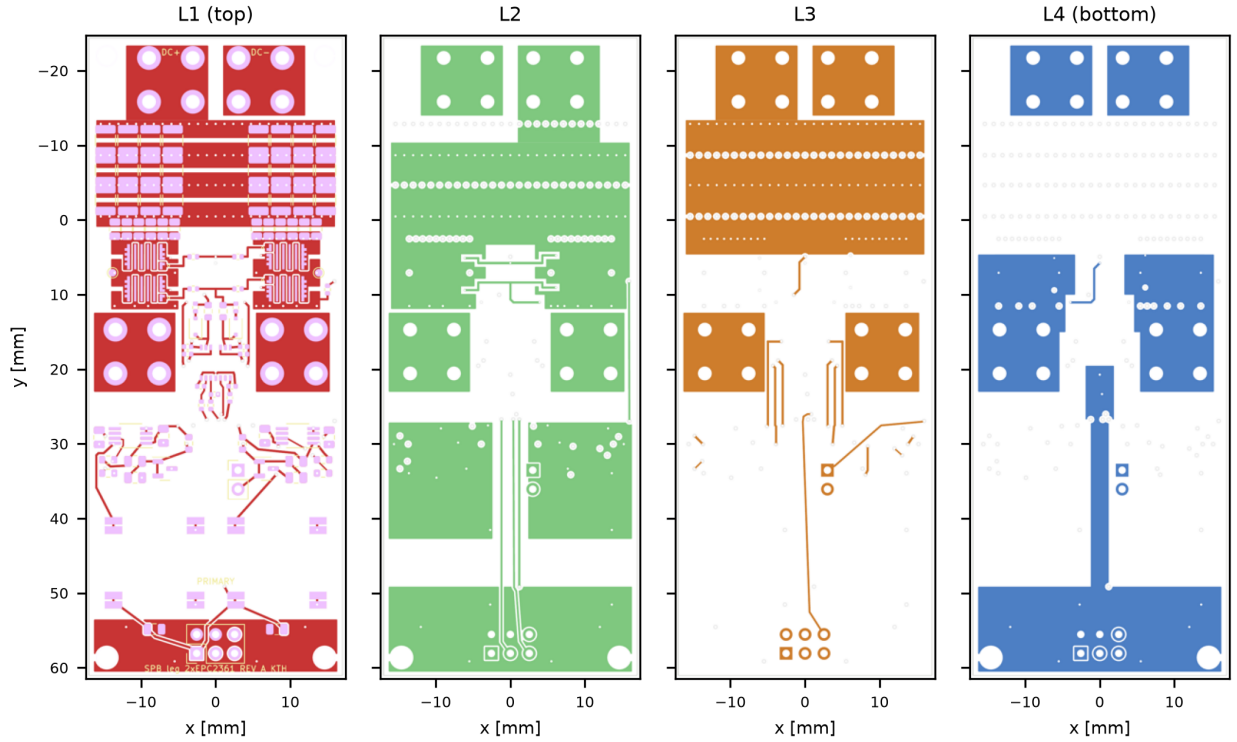


Figure 1: Copper layers of the Rev A board (top view on all layers). Red: L1, green: L2 (DC– return plane under the cells, VEE pours under the supply chains, GNDI at the bottom), orange: L3 (DC+ plane under the strips and the bank, supply columns), blue: L4 (switch-node plane, GNDI channel and bottom pour).

Table 1: Stack-up and design rules

Item	Value	Use
L1 (top), 70 μm	signal / power	FET and capacitor pads, strips, gate stars, all components
Prepreg L1–L2	0.1 mm	vertical commutation loop (sets L_{loop})
L2, 70 μm	DC– plane	loop return, Kelvin collectors in narrow keepouts, VEE pours on the tab
Core L2–L3	≈ 1.1 mm	1.6 mm total, symmetric build
L3, 70 μm	DC+ plane	feed to the cells; HS gate trunk and supply columns in the corridor
Prepreg L3–L4	0.1 mm	HS gate trunk (L3) over its Kelvin return (L4)
L4 (bottom), 70 μm	AC plane	switch node to J3/J4; HS Kelvin return under the trunk; GNDI
Min. track / clearance	0.15 / 0.15 mm	signal; power zones 0.2 mm clearance
Vias	0.6/0.3 mm (power, signal), 0.45/0.25 mm (Kelvin)	no via-in-pad, standard process
Finish, mask	ENIG, green LPI both sides	mask is the insulation for the 0.2 mm power gaps

The power-copper gaps of 0.2 mm between DC+, DC– and AC on L1 rely on the solder mask: IPC-2221B requires 0.13 mm for 51–100 V on coated external conductors (B4) and 0.1 mm on internal layers (B1), but 0.6 mm without coating (B2). The board therefore must not be used with exposed copper in the power area (no mask openings other than the pads).

2 Power transistors

Table 2: EPC2361 (datasheet, typical at 25 °C unless noted)

Parameter	Value	Note
$V_{DS} / V_{DS(tr)}$	100 V / 120 V	75 V submodule voltage (75 % utilisation)
$R_{DS(on)}$	0.75 m Ω (1.0 max)	$V_{GS} = 5$ V, 50 A
I_D cont. / pulsed	133 A / 697 A	
V_{GS} range	-4 ... +6 V	split rail +5.05/-1.24 V
$V_{GS(th)}$	0.8 / 1.1 / 2.5 V (min/typ/max)	bump ≈ 1.06 V at 0 V off, removed at -1.24 V
$C_{iss} / C_{oss} / C_{rss}$	3599 / 1019 / 13 pF	$V_{DS} = 50$ V
$Q_G / Q_{GS} / Q_{GD}$	28 / 8.5 / 3.8 nC	$V_{DS} = 50$ V, 50 A
Q_{oss}, Q_{RR}	90 nC, 0	
R_G internal	0.4 Ω	
$R_{\theta JC}$ (top) / $R_{\theta JB}$	0.2 / 1.5 K/W	top-side cooling through TIM to the cold plate
Package	3 $\times$ 5 mm, 7 stripes, 0.85 mm pitch	pin 1 gate, 2/4/6 source, 3/5/7 drain

The land pattern is the solder-mask-defined EPC pattern (hardware/epc2361-cell1). The drain stripes connect to the DC+ copper at one end of the package and the source stripes to the AC (high side) or DC- (low side) copper at the other end, as a comb on L1. The power vias are kept 0.3 mm outside the stripe ends (rows moved from 2.8 to 2.5 mm and from 11.1 to 11.5 mm). Between the high-side source and low-side drain stripes only 0.7 mm remains, which is too narrow for a via; the AC vias there were removed, and the switch node reaches L4 through the vias beside the packages. The package is chiral, so the left cell is rotated by 180°. The gate pads then lie at $y = y_{row} \pm 1.23$ mm, and the star on the symmetry line gives equal branch lengths.

3 Gate driver

3.1 Circuit

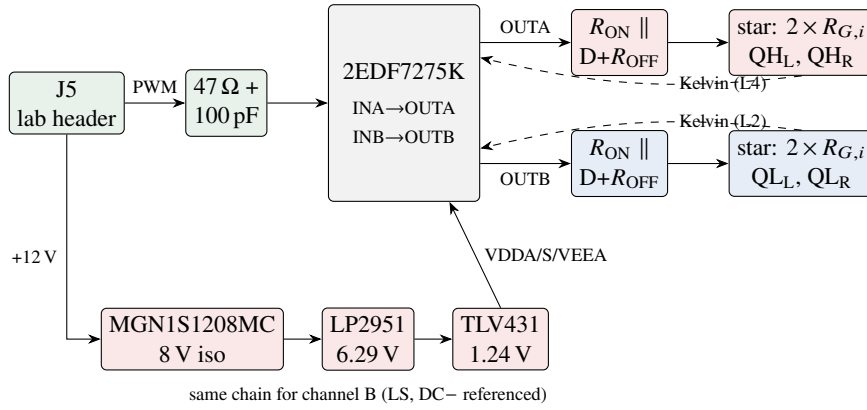


Figure 2: Gate-drive circuit of the leg. Colours mark the isolation domains: primary (green, GNDI), high side (red, switch-node referenced), low side (blue, DC- referenced).

The driver's input side runs from the 12 V lab supply in SLDO mode (SLDON to GNDI, $R_{VDDI} = 3$ k Ω , 22 nF at VDDI). Each PWM input has a 47 Ω /100 pF filter ($\tau = 4.7$ ns) against cable ringing; DISABLE is driven from the header (the driver's internal pull-down keeps it enabled when open). Channel A drives the high side and channel B the low side; dead time is generated by the controller.

3.2 Gate network: separate turn-on and turn-off paths

The 2EDF7275K has one output pin per channel, so separate turn-on and turn-off resistances need a steering diode. Each channel has

$$\text{OUT} \rightarrow (R_{ON} \parallel (D + R_{OFF})) \rightarrow \text{trunk} \rightarrow \text{star} \rightarrow R_{G,i} \rightarrow \text{gate}_i,$$

with the diode (PMEG3020EJ, SOD323F, cathode toward the driver output) conducting only when the driver pulls the gate down. Because the two transistors of a switch share the channel network, each transistor sees it twice:

$$R_{on,eff} = 2R_{ON} + R_{G,i}, \quad R_{off,eff} = 2(R_{ON} \parallel (R_{OFF} + r_D)) + R_{G,i} (+V_F),$$

plus $R_{G,int} = 0.4 \Omega$ and twice the driver output resistance (0.85Ω pull-up, 0.35Ω pull-down). The default population $R_{ON} = 0.75 \Omega$ (0603), $R_{OFF} = 0 \Omega$ (0402 jumper), $R_{G,i} = 0.5 \Omega$ (0402) gives 2.0Ω on and 0.5Ω plus the diode off per transistor, i.e. the simulated $R_{g,on}/R_{g,off} = 2/0 \Omega$ with the 0.5Ω star resistors kept for damping between the paralleled gates. The peak sink current is then 6.5 A per channel ($\approx 9.1 \text{ A}$ if also $R_{G,i} = 0$), inside the 8 A rating; R_{ON} , R_{OFF} and $R_{G,i}$ can be changed independently by population. Below the diode forward voltage ($\approx 0.4 \text{ V}$) the gate is pulled the rest of the way to -1.24 V through R_{ON} . The networks sit between the low-side star and the driver (Fig. 3); to make room the driver and the tab moved 3.5 mm down (board $34 \times 85.4 \text{ mm}$).

Table 3: Gate-drive budget per channel (two EPC2361, $V_{DD} - V_{EE} = 6.29 \text{ V}$)

	50 kHz	100 kHz
Gate charge per transistor, $Q_G(5 \text{ V}) + C_{iss} \cdot 1.24 \text{ V}$	32.5 nC	
Gate charge per channel	65 nC	
Peak gate current, turn-on / turn-off (default population)	3.1 A / 6.5 A	
Gate supply current $Q_{ch}f$	3.2 mA	6.5 mA
Gate drive power $Q_{ch}\Delta V f$	20 mW	41 mW
Total chain current (incl. 2 mA driver, 2.3 mA shunt bias)	7.6 mA	10.8 mA
LDO dissipation	13 mW	18 mW
MGN1 output power (125 mA, 1 W rating)	61 mW	87 mW

3.3 Decoupling

Each half rail of each channel has a $1 \mu\text{F}/16 \text{ V}$ X7R 0402 directly at the driver ($V_{DD}-S$ and $S-V_{EE}$), i.e. $139 \times$ the input capacitance of the two transistors (rule of thumb: $\geq 20 C_{iss} = 144 \text{ nF}$). A further $4.7 \mu\text{F}$ per half rail and $10 \mu\text{F}$ $V_{DD}-V_{EE}$ sit at the regulator, 10–12 mm away on the L3 supply columns, and supply the average current.

3.4 Gate-loop layout

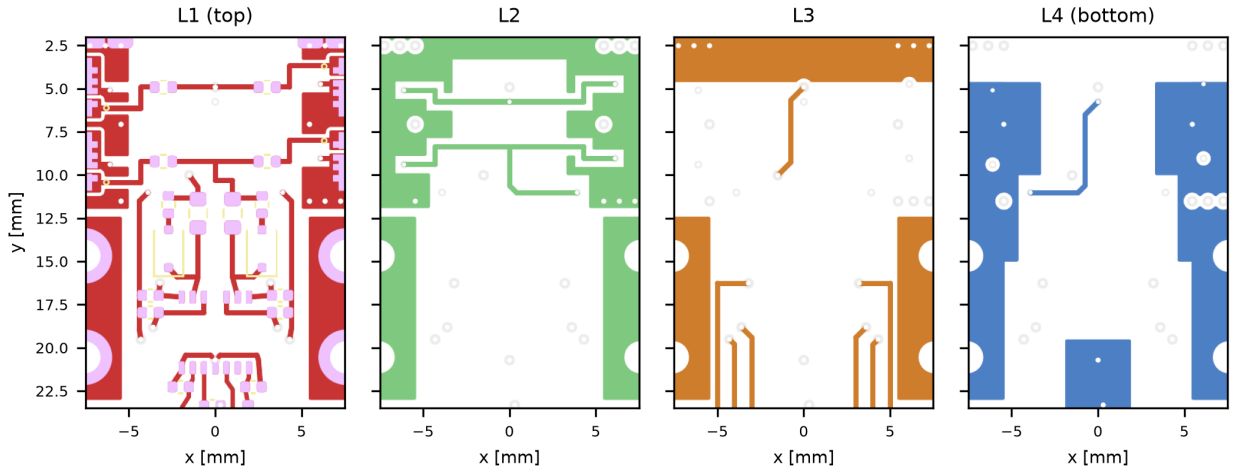


Figure 3: Driver corridor between the cells. L1: high-side star (upper row) and low-side star (lower row), each with two $R_{G,i}$; per channel R_{ON} (inner column) and R_{OFF} with the diode (outer column) above the 2EDF7275K; local decoupling to the left (A) and right (B) of the driver. L2: Kelvin collectors from the source-pad ends to the symmetry line in narrow keepouts of the DC– plane; the low-side collector continues to the B decoupling. L3: high-side trunk from the star via down the corridor; supply columns of both chains. L4: high-side Kelvin return directly under the L3 trunk inside the corridor keepout of the switch-node plane.

The extracted gate loop of the paper geometry is 1.71 nH per transistor with a common-source inductance of 0.030 nH (Q3D, Kelvin via at the source pad). The fabrication layout keeps the topology (star on the symmetry line, Kelvin at the source-pad end next to the gate) but lengthens the trunk into the wider corridor. The high-side pair (L3 trunk 0.1 mm above its L4 return) and the low-side pair (L1 trunk, L2 collector) are broadside-coupled, which keeps the loop area small, but their inductance must be re-extracted.

4 Isolated gate supplies

Each channel has its own MGN1S1208MC (12 V $\rightarrow$ 8 V, 125 mA, 2.5 pF isolation capacitance, CMTI > 200 kV/ μ s). Its unregulated 8 V output feeds an LP2951 in adjustable mode:

$$V_{DD} - V_{EE} = 1.235 \text{ V} \left(1 + \frac{102 \text{ k}\Omega}{24.9 \text{ k}\Omega} \right) = 6.29 \text{ V}.$$

A TLV431A between the source node S (cathode and reference) and VEE (anode) holds S at 1.24 V above VEE; 2.2 k Ω from VDD supplies the shunt bias (2.3 mA), so that $V_{DD} - V_S = +5.05 \text{ V}$ and $V_{EE} - V_S = -1.24 \text{ V}$. The driver's 4 V UVLO refers to $V_{DD} - V_{EE} = 6.29 \text{ V}$ and is not affected. The LP2951 (MSOP-8) replaces the earlier TLV1117 (SOT-223) because the chain must fit $14 \times 10 \text{ mm}$ next to the module; it needs only 0.4 V headroom and accepts the light-load rise of the MGN1 output (inputs rated 25 V, LP2951 30 V). Its SENSE and VTAP pins are left open and ERROR is not used; this and the output-capacitor stability with ceramics must be checked against the TI application circuit before ordering.

The MGN1 sits with its output pins toward the chain and its input pins toward the primary strip, so that its internal barrier (8.75 mm between the pin rows) separates the high-voltage and primary sides of the board. The VEE pours on L2 under each chain give the regulator a return plane in its own domain.

5 DC-link capacitors

The DC link is unchanged from the paper (Section V there): six 1 μ F 0805 per cell directly at the high-side drains define the commutation loop, and the $24 \times 10 \mu$ F 1210 bank supplies the ripple charge. The only fabrication change is the 0805 pitch of 1.6 mm (was 1.4 mm), so that neighbouring pads keep 0.15 mm.

Table 4: DC-link capacitors per leg (75 V, 50 kHz, rated point)

	local (per cell)	bank
Part	TDK C2012X7S2A105K125AB	Murata GRM32EC72A106KE05L
Size, dielectric	0805, X7S, 100 V	1210, X7S, 100 V
Number per leg	2×6	24 (3 rows of 8)
$C_{\text{nom}} / C_{\text{eff}}$ at 75 V*	1 μ F / 0.45 μ F	10 μ F / 3.0 μ F
ESR* / ESL* per part	6 m Ω / 0.48 nH	3 m Ω / 0.5 nH
Self-resonance	10.8 MHz	4.1 MHz
Ripple current per part	1.1 A _{rms}	2.0–2.1 A _{rms}
Loss per part	7.5 mW	11.5–13.3 mW

*assumed; to be confirmed with manufacturer models and impedance measurements

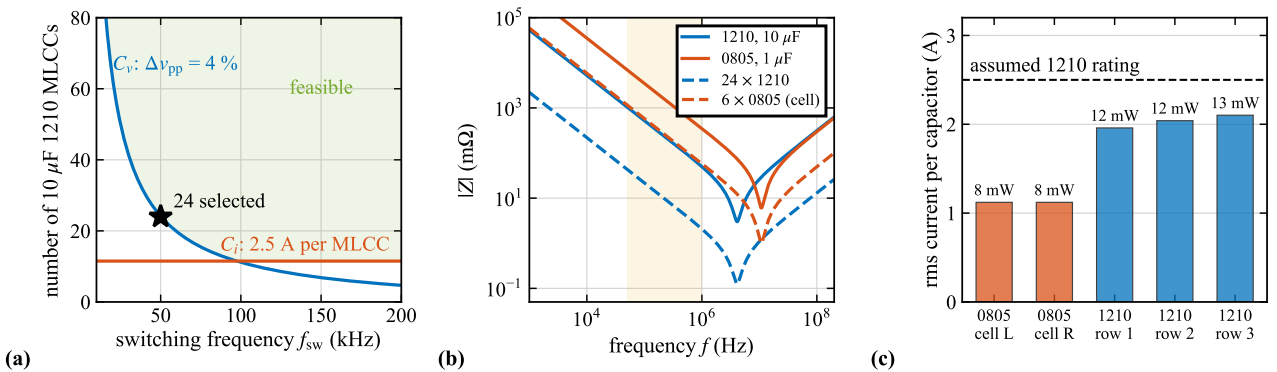


Figure 4: DC-link design (from the paper): (a) number of 1210 MLCCs set by voltage ripple and by ripple current, (b) impedance of parts, bank and local group, (c) rms current and loss per capacitor. Total capacitor loss per leg $\approx 0.39 \text{ W}$.

6 Parasitics

Table 5: Parasitics of the building block and of the cell busbar

Quantity	Source	Value
Power-loop inductance per cell	Q3D (paper geometry, 100 MHz)	0.374 nH
Gate loop L_G per transistor	Q3D BB_CS	1.71 nH
Common-source inductance, Kelvin / no Kelvin	Q3D BB_CS	0.030/0.061 nH
Mutual gate loop – gate loop	Q3D BB_CS	0.46 nH
ESL per MLCC, 0805 / 1210	assumed	0.48 / 0.5 nH
Busbar, leg to leg (A–B / A–C), 1 MHz	Q3D BB_BUS	3.35 / 4.37 nH
Busbar, leg to leg resistance A–B, 1 MHz	Q3D BB_BUS	1.757 m Ω
Busbar, series tab to leg (legs 1/2/3), 1 MHz	Q3D BB_BUS	41.12 / 41.28 / 41.35 nH
Busbar, series tab to leg, DC	Q3D BB_BUS	85.57 nH, 0.108 m Ω

The 0.2 mm gaps, via rows and plane outlines of the fabrication board follow the Q3D geometry, but 26 of the 168 power vias were removed for clearance to the FET stripes and the Kelvin traces, and the via rows were shifted. The 0.374 nH per cell is therefore a design value until the routed board is re-extracted (simulation/bb.q3d, import of the KiCad copper).

Busbar. The laminated cell busbar has 3.35 nH between neighbouring legs, which is what matters for the high-frequency current exchange between the three legs of a submodule (the local MLCCs close the commutation loops). The 41 nH from the series tab to a leg arise because the DC+ and DC– series tabs leave the cell at opposite ends and are not laminated there. This is acceptable for the low-frequency submodule current but should be reduced in the cell assembly by bringing both tabs out at the same end, or by a laminated cell-to-cell interconnect.

7 Connectors, isolation domains and I/O

Table 6: Connectors

Ref	Part	Domain	Function
J1, J2	Würth REDCUBE 74650174 (THR, M4)	DC+, DC–	submodule bus (busbar from below)
J3, J4	Würth REDCUBE 74650174 (THR, M4)	AC (switch node)	phase output, one per cell, directly below each switch node; connect both
J5	2 × 3 header, 2.54 mm	primary (GNDI)	1 +12 V, 2 GND, 3 PWM.H, 4 GND, 5 PWM.L, 6 DISABLE
J6	1 × 2 header, 2.54 mm	LS (DC–)	1 NTC (10 k, B3380, on the low-side source copper), 2 DC–. High-voltage domain: read out only through an isolated amplifier or with the DC link discharged
TP1, TP2	1 mm test pads	AC	switch-node probing
H1–H4	M2.5	–	insulated mounting (nylon or insulated sleeve)

Table 7: Minimum copper spacing between isolation domains on the routed board (pads, tracks and vias)

Domains	outer layers [mm]	inner layers [mm]
DC+ – HS (switch node)	0.20	1.71
DC+ – LS (DC–)	0.73	1.07
HS – LS	0.20	0.75
HS – primary	1.37	1.08
LS – primary	1.50	0.80

The power domains (DC+, switch node, DC–) differ by at most the submodule voltage (75 V nominal, 100 V rating) and are spaced for 100 V under solder mask. The primary side is separated from both gate-drive domains by at least 0.8 mm on inner and 1.37 mm on outer layers, and by 3.35 mm inside the 2EDF7275K package between its input and output pad rows. This is a *functional* separation for a lab board with a single submodule. In the stacked 1200 V inverter each submodule floats up to 1.2 kV against the controller; the PWM must then arrive through reinforced isolation (fibre optics or a reinforced digital isolator on the controller side), and the 12 V auxiliary supply must be isolated accordingly.

8 Fabrication data and BOM

All outputs are in `hardware/spb-bb-fab/fab/`:

- Gerber (L1–L4, masks, paste, silk, outline) and Gerber job file, Excellon drill file;
- schematic (`spb-bb-fab.kicad_sch`, PDF), pick-and-place (`spb-bb-fab-pos.csv`), assembly drawings top/bottom (PDF), STEP model;
- BOM (`spb-bb-fab-bom.csv`, 26 lines, 90 placed parts plus mounting holes and test pads).

The board is regenerated from `scripts/make_kicad_bb_fab.py` (placement, power copper from the Q3D geometry, hand-placed gate loops, a domain-restricted router `scripts/fab_router.py` for the supply and primary nets, power-via pruning, zone fill, DRC). The REDCUBE terminals are placed on the bottom side (press/solder from below); all other parts are on the top.

Table 8: Main BOM items (passives in the CSV)

Qty	Part	Function
4	EPC2361	power transistors
1	Infineon 2EDF7275K (PG-TFLGA-13-4)	dual isolated gate driver
2	Murata MGN1S1208MC-R7	isolated 12 V $\rightarrow$ 8 V, 1 W
2	TI LP2951 (VSSOP-8, adjustable)	6.29 V gate-rail regulator
2	TI TLV431AIDBZR	–1.24 V source-node shunt
12 / 24	TDK C2012X7S2A105K125AB / Murata GRM32EC72A106KE05L	local / bank DC link
4 / 2 / 2	$R_{G,i}$ 0.5 Ω 0402 / R_{ON} 0.75 Ω 0603 / R_{OFF} 0 Ω 0402	gate network (values by population)
2	Nexperia PMEG3020EJ (SOD323F)	turn-off steering diodes
4	Würth 74650174	DC and AC terminals
1	Murata NCP15XH103F03RC	NTC

9 Verification status and checklist

Done	DRC (0 violations, 0 unconnected); schematic netlist identical to the board (239 pins, <code>scripts/check_sch_vs_pcb.py</code>); domain spacing (Table 7); gate budget and peak currents (Table 3); footprints of the 2EDF7275K (datasheet Fig. 22) and MGN1 (KDC.MGN1 p. 17) entered from the datasheets; EPC land pattern from EPC.
Open, before ordering	Q3D extraction of the routed copper (power loop, gate loops, L_{CS}); LP2951 adjustable-mode pins and ceramic-capacitor stability; TLV431 DBZ pinout; footprint review of the custom LGA and MGN1 pads against the latest drawings; fab-house review of the 0.2 mm power gaps and the 0.45/0.25 mm Kelvin vias; thermal check of the AC path (two terminals, 4 layers 2 oz).
Bring-up	12 V only: check $V_{DD} - V_S$ and $V_{EE} - V_S$ on both channels; gate waveforms with the DC link at 0 V; double-pulse test at 24, 48, 75 V with a low-inductance probe on TP1/TP2; compare turn-on/off energy and the off-state gate bump with the simulation.